

Oja’s Algorithm for Streaming PCA: Tight Guarantees for Sparse Matrices

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Abstract

We study the behavior of Price & Xun’s modified version of Oja’s algorithm in the adversarial streaming PCA setting when the input matrix admits a sparse row representation. Given a matrix of the form $X = MQ^\top$ where Q is orthogonal and M has at most s nonzeros per row, we recover the top eigenvector using (near-)linear space when the spectral ratio satisfies $R = \Omega(s \log d)$. In particular, $R = \Omega(\log d)$ when $s = O(1)$. For the lower bound, we construct a hard instance on which Price & Xun’s algorithm fails for gaps $R = O(\log d)$, proving that the dependence on $\log d$ (and the constant-gap behavior for constant sparsity) is tight up to constants. Our analysis refines prior adversarial bounds by exploiting structural restrictions on per-row sparsity to obtain improved growth/rotation tradeoffs for the tracked vector.

1 Introduction

Principal Component Analysis is the problem of finding the k -dimensional subspace of the largest variance in a dataset. For large datasets, it is computationally expensive to keep track of the entire matrix $X \in \mathbb{R}^{n \times d}$, or the covariance matrix $\frac{1}{n}X^\top X \in \mathbb{R}^{d \times d}$. Our main objective is to analyze algorithms for PCA using only $\tilde{O}(d)$ space in the streaming setting, where data points arrive sequentially.

Oja’s algorithm begins with a random unit vector v_0 , iteratively updating with the rule $v_i = v_{i-1} + \eta x_i x_i^\top v_{i-1}$. While Oja’s algorithm performs well in practice, it has only been extensively studied in the stochastic setting, where data points are drawn i.i.d from a nice distribution.

Price & Xun [1] provide a modified version of Oja’s algorithm, analyzing it in the arbitrary data setting for the first time. So long as the spectral ratio fulfills $R = \lambda_1/\lambda_2 = \Omega(\log n \log d)$, this algorithm can be shown to converge to the top principal component. Kacham & Woodruff [2] combine the modified Oja’s algorithm with row-norm sketching, randomly selecting $n = \text{poly}(d)$ rows in the stream to improve the spectral ratio requirement of arbitrary streams to $R = \Omega(\log^2 d)$. This analysis also gives a lower bound of $R = O(\log d / \log \log d)$ for the modified Oja’s algorithm to converge.

In this work, we improve the analysis of Price & Xun for matrices of the form $X = MQ^\top$ where $Q \in \mathbb{R}^{d \times d}$ is orthogonal and $M \in \mathbb{R}^{n \times d}$ has at most s nonzero entries per row for

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some sparsity coefficient s . We show the modified version of Oja’s algorithm works on such inputs given $R = \Omega(s \log d)$ for some sparsity coefficient s . We then improve the lower bound by showing the algorithm fails for gaps $R = O(\log d)$. Our proof consists of a hard instance that is a sparse stream with only $O(1)$ many nonzero entries per row, showing the spectral gap requirement for sparse matrices where $s = O(1)$ is tight up to a constant multiplicative factor.

1.1 Related Work

We build on recent analyses of Oja’s algorithm in the adversarial streaming setting, which extends from a broad line of low-memory eigenvector approximation and matrix sketching algorithms. We follow the technical framework & notation introduced by Price & Xun [1]. In this work, they show that a modified Oja’s algorithm for arbitrary data outputs a suitable estimate for the top eigenvector, converging when $R = \Omega(\log n \log d)$.

Kacham & Woodruff [2] improved this algorithm, allowing it to succeed for $R = \Omega(\log^2 d)$. Their algorithm uses row-norm sketching to guarantee $n = \text{poly}(d)$. They also extend the analysis of Price & Xun’s beyond the worst-case setting, noting that a spectral gap of $R = \Omega(\log d)$ is sufficient when the rows are provided in a random order. Independently, they also provide a new algorithm for eigenvector approximation in uniform order streams.

A broader class of low-memory approaches target top-eigenvector computation in the streaming setting. Notably, FrequentDirections for adversarial PCA maintains a $O(dk)$ “sketch” matrix that yields a k -rank approximation with deterministic error guarantees linear in k ([3], [4]). A key difference between the analysis of FrequentDirections and our analysis of Oja’s algorithm is the error metric: we use sin-squared error with conditions on the spectral ratio, whereas a Frobenius guarantee is generally more common for low-rank approximations.

2 Upper Bound

We first provide a brief summary of Price & Xun’s upper bound, which centers around showing the growth of the vector being tracked throughout the algorithm $\|v_n\|$ is large. The main technical challenge arises in bounding the quantity $\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2$. Naively, one can bound this by $\sigma_2^2(d-1)$, which is tight if $P\hat{v}_{i-1}$ rotates through all $d-1$ nonprincipal orthogonal directions. The main technical insight, Lemma 2.5, states that, given $v_0 = v_*$,

$$\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 \lesssim \sigma_2^2 \log^2 n \log \|v_n\|$$

Combined with Lemma 2.6, one gets that

$$\begin{aligned} \log \|v_n\|^2 &\geq \frac{1}{4}\sigma_1 - \eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 \geq \frac{1}{4}\sigma_1 - \sigma_2^2 \log^2 n \log \|v_n\| \\ \therefore \log \|v_n\| &\gtrsim \frac{\sigma_1}{1 + \sigma_2^2 \log^2 n} \end{aligned} \tag{1}$$

Finally, Claim 3.3 and Lemma 2.1 allow us to obtain

$$\|P\hat{v}_n\| \leq \sqrt{\sigma_2} + \frac{\|Pv_0\|}{\|v_n\|}$$

The assumption that $v_0 = v_*$ leverages the linearity of Oja's algorithm, allowing us to hone in on the v_* component of our initial random vector. The downside is that since the expected magnitude of this component is $O(1/\sqrt{d})$, we must make up for an additional $\text{poly}(d)$ factor of growth. But for our vector to grow polynomially in d , we must have that $\log \|v_n\| \simeq \sigma_1$, as $\sigma_1 = \Omega(\log d)$ by assumption. In equation (1), this requires $\sigma_2 = O(1/\log n)$, and hence $R = \Omega(\log n \log d)$.

We observe that $\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2$ cannot be large without $\hat{v}_{i-1}$ rotating into many distinct directions, as variance in any fixed direction is bounded by σ_2 . Indeed, the proof of Lemma 2.5 in [1] heavily relies on Lemma 2.2, which shows that $\hat{v}_{i-1}$ cannot rotate much without increasing its norm. We double down on this intuition by noting that for sparse matrices, the rotation of v_{i-1} at each iteration is not particularly complex, and the relationship between growth and rotation can be made much more explicit.

First, when each data point lies in the span of v_* and exactly 1 nonprincipal component, we improve the analysis to show $\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 \lesssim \sigma_2^2 \log \|v_n\|$ in Lemma 2.2. We then generalize this in Lemma 2.3 to streams where each data point is a linear combination of v_* and s orthogonal directions. This directly implies $R = \Omega(\log d)$ suffices when $s = O(1)$.

Theorem 2.1 (Sparse matrix guarantees). *Suppose $X \in \mathbb{R}^{n \times d}$ with spectral gap $R = \Omega(\log d)$, where $X = MQ^\top$ for $M \in \mathbb{R}^{n \times d}$ has $O(1)$ nonzero entries per row, and $Q \in \mathbb{R}^{d \times d}$ is orthogonal. Then there exists an algorithm using $\tilde{O}(d)$ space that approximates the top principal component obtaining $O(\frac{\log d}{R} + \frac{1}{\text{poly}(d)})$ error with high probability.*

2.1 Bound for sparsity of 1

We fix an arbitrary basis $w_1, \dots, w_{d-1}$ spanning the subspace orthogonal to v_* . Let $S_j = \{i : \langle x_i, w_j \rangle \neq 0\}$ be the set of indices with nonzero contribution in the j th orthogonal direction, for each $j \in [d-1]$. Furthermore, let $h(i) : [n] \rightarrow 2^{[d]}$ map each index i to the set of orthogonal directions contributing to x_i ; that is, $h(i) = \{j : \langle x_i, w_j \rangle \neq 0\}$.

Lemma 2.2. *Given $|h(i)| \leq 1$ for all $i \in [n]$,*

$$\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 \leq \sigma_2^2 \log \|v_n\|$$

Proof. As each x_i only contributes to 1 orthogonal direction, $w_{h(i)}$, we can rewrite $\langle x_i, P\hat{v}_{i-1} \rangle^2 = \langle x_i, w_{h(i)} \rangle^2 \langle \hat{v}_{i-1}, w_{h(i)} \rangle^2$, which gives us

$$\begin{aligned} \eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 &= \eta \sum_{i=1}^n \langle x_i, w_{h(i)} \rangle^2 \langle \hat{v}_{i-1}, w_{h(i)} \rangle^2 \\ &\leq \eta \sum_{i=1}^n \langle x_i, w_{h(i)} \rangle^2 \frac{\left(\eta \sum_{k=1}^{i-1} \langle x_k, w_{h(i)} \rangle \langle x_k, v_{k-1} \rangle \right)^2}{1 + 2\eta \sum_{k=1}^{i-1} \langle x_k, v_{k-1} \rangle^2} \\ &\leq \eta \sum_{i=1}^n \langle x_i, w_{h(i)} \rangle^2 \frac{\left(\eta \sum_{\substack{k < i \\ x_k \in S_{h(i)}}} \langle x_k, w_{h(i)} \rangle \langle x_k, v_{k-1} \rangle \right)^2}{1 + 2\eta \sum_{k=1}^{i-1} \langle x_k, v_{k-1} \rangle^2} \end{aligned}$$

where the denominator arises from the bound on $\|v_{i-1}\|^2$ as per Lemma 4.1. Using Cauchy-Schwarz on the numerator, we get

$$\begin{aligned}
&\leq \eta \sum_{i=1}^n \langle x_i, w_{h(i)} \rangle^2 \left(\eta \sum_{\substack{k < i \\ x_k \in S_{h(i)}}} \langle x_k, w_{h(i)} \rangle^2 \right) \frac{\eta \sum_{\substack{k < i \\ x_k \in S_{h(i)}}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2} \\
&\leq \sigma_2 \eta \sum_{i=1}^n \langle x_i, w_{h(i)} \rangle^2 \frac{\eta \sum_{\substack{k < i \\ x_k \in S_{h(i)}}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2} \\
&= \sigma_2 \eta \sum_{j=1}^{d-1} \sum_{x_i \in S_j} \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{k < i \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}$$

The fraction represents the ratio of growth between the points contributing to orthogonal direction j and the total growth up until index i . Define i_j^* to be the index corresponding to the maximum "growth ratio" obtained by direction j , i.e.

$$i_j^* = \arg \max_{i \in S_j} \frac{\eta \sum_{\substack{k < i \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2}$$

Thus, our summation is bounded above by

$$\begin{aligned}
\sigma_2 \eta \sum_{j=1}^{d-1} \sum_{x_i \in S_j} \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{k < i_j^* \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} &= \sigma_2 \eta \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{k < i_j^* \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} \sum_{x_i \in S_j} \langle x_i, w_j \rangle^2 \\
&\leq \sigma_2^2 \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{k < i_j^* \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}$$

Necessarily, $i_j^* \in S_j$. As each S_j is disjoint, we may assume the indices i_j^* are distinct. Without loss of generality, we permute the orthogonal directions in increasing order of i_j^* , so $i_{(1)}^* < \dots < i_{(d-1)}^*$. Then, observe that

$$\frac{\eta \sum_{\substack{k < i_j^* \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} = 1 - \left(\frac{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_j}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} \right)$$

Let $S_{>j} = \bigcup_{j' > j} S_{j'}$. We observe $\frac{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_j}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} \leq 1$, so subtracting $\eta \sum_{\substack{k < i_j^* \\ k \notin S_j}} \langle x_k, v_{k-1} \rangle^2$ from both the numerator and denominator decreases it. Thus,

$$1 - \left(\frac{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_j}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} \right) \leq 1 - \left(\frac{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_{>j-1}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_{>j}}} \langle x_k, v_{k-1} \rangle^2} \right)$$

$$\begin{aligned}
&\leq \log \left(\frac{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_{>j}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_{>j-1}}} \langle x_k, v_{k-1} \rangle^2} \right) \\
&\leq \log \left(\frac{1/2 + \eta \sum_{\substack{k < i_{(j+1)}^* \\ k \notin S_{>j}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_{>j-1}}} \langle x_k, v_{k-1} \rangle^2} \right)
\end{aligned}$$

where the second inequality uses that $1 - \frac{1}{x} \leq \log x$, $x \in \mathbb{R}$. Thus,

$$\begin{aligned}
\sigma_2^2 \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{k < i_j^* \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} &\leq \frac{1}{2} \sigma_2^2 \sum_{j=1}^{d-1} \log \left(\frac{1/2 + \eta \sum_{\substack{k < i_{(j+1)}^* \\ k \notin S_{>j}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_j^* \\ k \notin S_{>j-1}}} \langle x_k, v_{k-1} \rangle^2} \right) \\
&= \frac{1}{2} \sigma_2^2 \log \frac{1/2 + \eta \sum_{k < i_{d-1}^*} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_1^* \\ k \notin S_{>1}}} \langle x_k, v_{k-1} \rangle^2} \\
&\leq \frac{1}{2} \sigma_2^2 \log(1 + 2\eta \sum_{k=1}^n \langle x_k, v_{k-1} \rangle^2) \leq \frac{1}{2} \sigma_2^2 \log \|v_n\|^2
\end{aligned}$$

so $\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 \leq \sigma_2^2 \log \|v_n\|$, as desired. $\square$

2.2 Generalizing to sparsities of s

We now generalize the result to a sparsity of s . The main challenge here is that the indices i_j^* can no longer be assumed to be disjoint. We resolve this by considering s disjoint copies of growth ratio indices, though this ultimately loosens the upper bound by a factor of s^2 .

Lemma 2.3. *Given $|h(i)| \leq s$ for all $i \in [n]$,*

$$\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 \leq s^2 \sigma_2^2 \log \|v_n\|$$

Proof. Observe that

$$\eta \sum_{i=1}^n \langle x_i, P\hat{v}_{i-1} \rangle^2 = \eta \sum_{i=1}^n \left(\sum_{j=1}^{d-1} \langle x_i, w_j \rangle \langle \hat{v}_{i-1}, w_j \rangle \right)^2 = \eta \sum_{i=1}^n \left(\sum_{j \in h(i)} \langle x_i, w_j \rangle \langle \hat{v}_{i-1}, w_j \rangle \right)^2$$

We know $|h(i)| \leq s$, so by Cauchy-Schwarz on the inner summation,

$$\begin{aligned}
\eta \sum_{i=1}^n \left(\sum_{j \in h(i)} \langle x_i, w_j \rangle \langle \hat{v}_{i-1}, w_j \rangle \right)^2 &\leq \eta \sum_{i=1}^n |h(i)| \sum_{j \in h(i)} \langle x_i, w_j \rangle^2 \langle \hat{v}_{i-1}, w_j \rangle^2 \\
&\leq s\eta \sum_{i=1}^n \sum_{j \in h(i)} \langle x_i, w_j \rangle^2 \frac{\left(\eta \sum_{\substack{k \in S_j \\ k < i}} \langle x_k, v_{k-1} \rangle \langle x_k, w_j \rangle \right)^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}$$

$$\begin{aligned}
&\leq s\eta \sum_{i=1}^n \sum_{j \in h(i)} \langle x_i, w_j \rangle^2 \eta \sum_{\substack{x_k \in S_j \\ k < i}} \langle x_k, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k < i}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2} \\
&\leq s\eta\sigma_2 \sum_{i=1}^n \sum_{j \in h(i)} \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k < i}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2} \\
&\leq s\eta\sigma_2 \sum_{j=1}^{d-1} \sum_{i=1}^n \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k < i_j^*}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} \\
&\leq s\sigma_2^2 \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{x_k \in S_j \\ k < i_j^*}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}$$

Arbitrarily ordering each set $h(i)$ and define $S_{j,l} \subseteq S_j$ as the set of all indices i for which $\langle x_i, w_j \rangle \neq 0$ and x_i appears as the l th index in $h(i)$ (for $1 \leq l \leq s$, where $h(i)_l = 0$ if $l > |h(i)|$). We then define $i_{j,l}^*$ as the index obtaining the maximum growth ratio constrained to the indices in $S_{j,l}$; that is,

$$i_{j,l}^* = \arg \max_{i \in S_{j,l}} \frac{\eta \sum_{\substack{k < i \\ x_k \in S_{j,l}}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i} \langle x_k, v_{k-1} \rangle^2}$$

As $k \in S_j \implies k \in S_{j,l}$ for some l ,

$$\begin{aligned}
s\sigma_2^2 \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{k < i_j^* \\ x_k \in S_j}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_j^*} \langle x_k, v_{k-1} \rangle^2} &= s\sigma_2^2 \sum_{j=1}^{d-1} \sum_{l=1}^s \frac{\eta \sum_{\substack{k < i_{j,l}^* \\ x_k \in S_{j,l}}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_{j,l}^*} \langle x_k, v_{k-1} \rangle^2} \\
&\leq s\sigma_2^2 \sum_{j=1}^{d-1} \sum_{l=1}^s \frac{\eta \sum_{\substack{k < i_{j,l}^* \\ k \in S_{j,l}}} \langle x_k, v_{k-1} \rangle^2}{1 + 2\eta \sum_{k < i_{j,l}^*} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}$$

Now, for any $l \in [s]$, $S_{j,l} \cap S_{j',l} = \emptyset$ for any pair $j \neq j'$, so the indices $i_{j,l}^*$ are disjoint for a fixed l . Denoting $S_{>j,l} = \bigcup_{j' > j} S_{j',l}$, we may proceed as in Lemma 2.2:

$$\begin{aligned}
&\leq s\sigma_2^2 \sum_{l=1}^s \frac{1}{2} \sum_{j=1}^{d-1} \left(1 - \frac{1/2 + \eta \sum_{\substack{k < i_{j,l}^* \\ k \notin S_{j,l}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{k < i_{j,l}^*} \langle x_k, v_{k-1} \rangle^2} \right) \\
&\leq s\sigma_2^2 \sum_{l=1}^s \frac{1}{2} \sum_{j=1}^{d-1} \left(1 - \frac{1/2 + \eta \sum_{\substack{k < i_{j,l}^* \\ k \notin S_{>j-1,l}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_{j,l}^* \\ k \notin S_{>j,l}}} \langle x_k, v_{k-1} \rangle^2} \right) \\
&\leq s\sigma_2^2 \sum_{l=1}^s \frac{1}{2} \sum_{j=1}^{d-1} \log \left(\frac{1/2 + \eta \sum_{\substack{k < i_{j,l}^* \\ k \notin S_{>j,l}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_{j,l}^* \\ k \notin S_{>j-1,l}}} \langle x_k, v_{k-1} \rangle^2} \right)
\end{aligned}$$

$$\begin{aligned}
&\leq s\sigma_2^2 \sum_{l=1}^s \frac{1}{2} \sum_{j=1}^{d-1} \log \left(\frac{1/2 + \eta \sum_{\substack{k < i_{j+1,l}^* \\ k \notin S_{>j,l}}} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{\substack{k < i_{j,l}^* \\ k \notin S_{>j-1,l}}} \langle x_k, v_{k-1} \rangle^2} \right) \\
&\leq s\sigma_2^2 \sum_{l=1}^s \frac{1}{2} \log \frac{1/2 + \eta \sum_{k < i_{d-1}^*} \langle x_k, v_{k-1} \rangle^2}{1/2 + \eta \sum_{k < i_{j,1}^*}} \\
&\leq s\sigma_2^2 \frac{1}{2} \log \|v_n\|^2 \leq s^2 \sigma_2^2 \log \|v_n\|
\end{aligned}$$

as desired. $\square$

Finally, we may prove Theorem 2.1 by $s = O(1)$ in Lemma 2.3, giving us an improved spectral upper bound for sparse matrices.

Proof of Theorem 2.1. Substituting our bound from Lemma 2.3 into Lemma 2.6 of [1],

$$\begin{aligned}
\log \|v_n\| &\geq \frac{1}{4} \sigma_1 - s^2 \sigma_2^2 \log \|v_n\| \\
\log \|v_n\| &\geq \frac{\sigma_1}{4(1 + s^2 \sigma_2^2)}
\end{aligned}$$

For $\sigma_2 = O(\frac{1}{s}) = O(1)$, we have that $\log \|v_n\| \geq \sigma_1 = \Omega(\log d)$, so by Theorem 1.2 of [1], the modified version of Oja's algorithm outputs $\hat{v}_n$ satisfying $1 - |\langle v_n, v_* \rangle|^2 \leq O(\frac{\log d}{R} + \frac{1}{\text{poly}(d)})$ with high probability. $\square$

Note that one may obtain the more general bound of $R = \Omega(s \log d)$ for matrices with row sparsity s . This bound is better than the upper bound provided in [2] so long as $s = O(\log d)$. We leave as an open problem whether the spectral requirement for modified Oja's can be improved for adversarial streams.

3 Lower Bound

We first provide a general Gaussian anti-concentration result.

Lemma 3.1. *For $X \sim \mathcal{N}(0, \sigma^2)$ and $\delta > 0$, $\Pr(|X| \geq \delta \sigma \sqrt{\pi/2}) \geq 1 - \delta$.*

Proof. Observe $|X|/\sigma \sim \mathcal{N}(0, 1)$. As the standard Gaussian density is bounded by $1/\sqrt{2\pi}$, $\Pr(|X|/\sigma \leq \delta \sqrt{\pi/2}) \leq \delta$. Rewriting the condition as $|X| \leq \delta \sigma \sqrt{\pi/2}$ and taking the complement gives us the desired result. $\square$

Next, we provide an instance for which Oja's algorithm fails to provide a satisfactory output for $R = O(\log d)$. Similar to the analysis of [2], we show that, given necessary constraints on η , the non-principal components grow a constant factor more than the principal component.

Theorem 3.2. *For a fixed constant $C > 0$ and any $R < C \log d$, there exists a matrix $X \in \mathbb{R}^{n \times d}$ with spectral ratio at least $R/3$, $n = O(R)$, and principal component v_* on which Oja's algorithm outputs $\hat{v}_n$ satisfying $|\langle \hat{v}_n, v_* \rangle| \leq 3C\sqrt{2}$ with probability at least 0.9. Additionally, the largest norm row in X is an insufficient estimate of v_* .*

Proof. The $n = R + 5$ rows of our instance are as follows:

$$\begin{aligned} x_1 &= \frac{1}{\sqrt{R}}(e_1 - e_2), & x_2 &= \frac{1}{\sqrt{R}}(e_2 - e_3) \\ x_i &= \frac{1}{\sqrt{R}}e_1 & \forall i \in [3, R+1] \\ x_{R+2} &= \frac{1}{\sqrt{R}}(e_1 + e_2), & x_{R+3} &= \frac{1}{\sqrt{R}}(e_2 + e_3), & x_{R+4} &= \frac{1}{\sqrt{R}}e_3, & x_{R+5} &= \sqrt{\frac{3}{R}}e_4 \end{aligned}$$

We first consider the principal component of our data stream $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$. Denote the j th

column of X to be c_j for $j \in [d]$, and note $c_j = \begin{pmatrix} \langle x_1, e_j \rangle \\ \langle x_2, e_j \rangle \\ \vdots \\ \langle x_n, e_j \rangle \end{pmatrix}$. By construction, $c_{j_1}^\top c_{j_2} =$

$\sum_{k=1}^d \langle x_k, e_{j_1} \rangle \langle x_k, e_{j_2} \rangle = 0$ whenever $j_1 \neq j_2$. As $X^\top X$'s off-diagonal terms correspond to the values $c_{j_1}^\top c_{j_2}$ for $j_1 \neq j_2$, we conclude $X^\top X$ is a diagonal matrix, and thus its eigenvectors are $e_1, e_2, \dots, e_d$ in some order. We note the nonzero eigenvalues are $1 + 1/R$ corresponding to e_1 , $3/R$ for e_2, e_3 , & e_4 , and 0 for e_j , $2 \leq j \leq d$. Thus, the spectral ratio of X is $(1 + 1/R)/(3/R) \geq R/3$. We also note the maximum magnitude data point is $x_{R+5} = e_4$, which is orthogonal to the principal component and hence wouldn't be an acceptable estimate.

If $C \geq \frac{1}{3\sqrt{2}}$, the theorem holds vacuously. Otherwise, assume for contradiction that $|\langle \hat{v}_n, e_1 \rangle| > c$ with probability ≥ 0.1 , where $c = 3C\sqrt{2} < 1$. The initial vector $\hat{v}_0$ may be expressed as $v_0/\|v_0\|$, where $v_0 \sim \mathcal{N}(0, I_d)$. We first note that $\Pr(\|v_0\|^2 \leq d/4) \leq e^{-d/8}$, so w.p. $\geq 1 - e^{-d/8}$, $\|v_0\| > \sqrt{d}/2$.

We now show η/R must be large for $|\langle \hat{v}_n, e_1 \rangle| > c$ to hold. As $\langle v_n, e_1 \rangle$ can be written as a linear combination of independent standard Gaussians $\langle v_0, e_1 \rangle, \langle v_0, e_2 \rangle, \langle v_0, e_3 \rangle$, we get that $\langle v_n, e_1 \rangle \sim \mathcal{N}(0, s^2)$, where

$$\begin{aligned} s^2 &= \left(\left(1 + \frac{\eta}{R}\right)^{R+1} - \left(\frac{\eta}{R}\right)^2 \left(1 + \frac{\eta}{R}\right) \right)^2 + \left(\frac{\eta}{R} \left(\left(1 + \frac{\eta}{R}\right)^R + \left(1 + \frac{\eta}{R}\right)^2 \right) \right)^2 + \left(\frac{\eta}{R}\right)^4 \\ &\leq 3 \left(1 + \frac{\eta}{R}\right)^{2R+2} \end{aligned}$$

We have that $c < |\langle \hat{v}_n, e_1 \rangle| \leq \frac{|\langle v_n, e_1 \rangle|}{\|v_0\|}$ or equivalently $|\langle v_n, e_1 \rangle| \geq c\|v_0\| \geq \frac{c\sqrt{d}}{2}$ w.h.p. By a Gaussian tail bound,

$$\Pr \left(|\langle v_n, e_1 \rangle| \geq \frac{c\sqrt{d}}{2} \right) \leq 2 \exp \left(-\frac{dc^2}{8s^2} \right) \leq 2 \exp \left(-\frac{dc^2}{24(1 + \eta/R)^{2R+2}} \right)$$

By assumption, this probability is at least $1/d$, so necessarily $\exp \left(\frac{dc^2}{24(1 + \eta/R)^{2R+2}} \right) \leq 2d$, or equivalently $\eta \geq R \left(\left(\frac{dc^2}{24 \log \log 2d} \right)^{\frac{1}{2R+2}} - 1 \right) \geq \frac{R}{2R+2} (\log d + \log \frac{c^2}{24 \log 2d})$, where the second inequality uses that $e^x - 1 \geq x$ for $x \in \mathbb{R}$. Using that $R < C \log d$, we see that $\frac{\eta}{R} \geq \frac{1}{3C}$ w.h.p.

Given this lower bound on η/R , we next show that the growth in the non-principal directions is significant. First, note that

$$\langle v_n, e_2 \rangle = \frac{\eta}{R} \left(1 + \frac{\eta}{R}\right)^R \langle v_1, e_1 \rangle + \left(1 + \frac{\eta}{R}\right)^2 \langle v_2, e_2 \rangle + \frac{\eta}{R} \langle v_2, e_3 \rangle$$

where $\langle v_2, e_2 \rangle \sim \mathcal{N}(0, t^2)$, and $\langle v_2, e_3 \rangle \sim \mathcal{N}(0, u^2)$, where $t^2 = \left(\frac{\eta}{R}\right)^2 + \left(\frac{\eta}{R}\right)^2(1 + \frac{\eta}{R})^2 + (1 + \frac{\eta}{R})^4$, $u^2 = \left(\frac{\eta}{R}\right)^4 + \left(\frac{\eta}{R}\right)^2(1 + \frac{\eta}{R})^2 + (1 + \frac{\eta}{R})^2$. Note that $\langle v_1, e_1 \rangle$ is a Gaussian with variance at least 1, so by Lemma 3.1, $|\langle v_1, e_1 \rangle| \geq \frac{1}{10}$ w.p. $\geq 1/11$. As $t^2, u^2 \leq 3(1 + \frac{\eta}{R})^4$,

$$\Pr \left(\left(1 + \frac{\eta}{R}\right)^2 |\langle v_n, e_2 \rangle| \geq \frac{\epsilon}{20} \frac{\eta}{R} \left(1 + \frac{\eta}{R}\right)^R \right) \leq 2 \exp \left(-\frac{\epsilon^2 (\eta/R)^2 (1 + \eta/R)^{2R-4}}{800t^2} \right) \leq 2 \exp(-\text{poly}(d)/\epsilon^2)$$

for any fixed $\epsilon > 0$. Similarly, $\Pr \left(\frac{\eta}{R} |\langle v_n, e_2 \rangle| \geq \frac{\epsilon}{20} \frac{\eta}{R} \left(1 + \frac{\eta}{R}\right)^R \right) \leq 2 \exp(-\text{poly}(d)/\epsilon^2)$, so $|\langle v_n, e_2 \rangle| \geq (1 - \epsilon) \frac{\eta}{R} \left(1 + \frac{\eta}{R}\right)^R |\langle v_1, e_1 \rangle|$ w.h.p. Furthermore,

$$\langle v_n, e_3 \rangle = \left(\frac{\eta}{R}\right)^2 \left(1 + \frac{\eta}{R}\right)^R \langle v_1, e_1 \rangle + \frac{\eta}{R} \left(1 + \frac{\eta}{R}\right)^2 \langle v_2, e_2 \rangle + \left(1 + \frac{\eta}{R}\right)^2 \langle v_2, e_3 \rangle$$

and we similarly obtain $|\langle v_n, e_2 \rangle| \geq (1 - \epsilon) \frac{\eta}{R} (1 + \frac{\eta}{R})^R |\langle v_1, e_1 \rangle|$ with probability $\geq 1 - 4 \exp(-\text{poly}(d)/\epsilon^2)$.

Finally, $\langle v_n, e_1 \rangle = (1 + \frac{\eta}{R})^R \langle v_1, e_1 \rangle + \frac{\eta}{R} \langle v_2, e_2 \rangle$, and so $|\langle v_n, e_1 \rangle| \leq (1 + \epsilon)(1 + \frac{\eta}{R})^R |\langle v_1, e_1 \rangle|$ w.p. $\geq 1 - 2 \exp(-\text{poly}(d)/\epsilon^2)$ as well.

Since $\|Pv_n\|^2 \geq \langle v_n, e_2 \rangle^2 + \langle v_n, e_3 \rangle^2$, we conclude that

$$c^2 < \langle \hat{v}_n, e_1 \rangle^2 \leq \frac{\langle v_n, e_1 \rangle^2}{\langle v_n, e_2 \rangle^2 + \langle v_n, e_3 \rangle^2} \leq \frac{2}{(\eta/R)^2 + (\eta/R)^4}$$

for sufficiently small ϵ , and with high probability. But since $\frac{\eta}{R} \geq \frac{1}{3C}$, we obtain

$$\frac{1}{9C^2} \leq \left(\frac{\eta}{R}\right)^2 < \frac{2}{c^2} = \frac{1}{9C^2}$$

with probability $\geq 1 - \frac{1}{11} - O(1/d) > 9/10$, which is a contradiction. $\square$

References

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4 Appendix

Lemma 4.1. *For $i \in [1, n]$,*

$$||v_i||^2 \geq ||v_0||^2 + 2\eta \sum_{k=1}^i \langle x_k, v_{k-1} \rangle^2$$

Proof. By the Oja update rule,

$$\begin{aligned} ||v_i||^2 &= \langle v_{i-1} + \eta \langle x_i, v_{i-1} \rangle x_i, v_{i-1} + \eta \langle x_i, v_{i-1} \rangle x_i \rangle \\ &= ||v_{i-1}||^2 + 2\eta \langle x_i, v_{i-1} \rangle^2 + 2\eta^2 ||x_i||^2 \langle x_i, v_{i-1} \rangle^2 \\ &\geq ||v_{i-1}||^2 + 2\eta \langle x_i, v_{i-1} \rangle^2 \end{aligned}$$

Expanding recursively gives us $||v_i||^2 \geq ||v_0||^2 + 2\eta \sum_{k=1}^i \langle x_k, v_{k-1} \rangle^2$, as desired. $\square$